Khulna University of Engineering & Technology

ECE 3200: Electronics Project Design/ Development

Project Title: Automated Car Parking System

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Introduction

1.1 Background and Motivation

In modern urban environments, efficient parking management has become increasingly critical due to rising vehicle density. Traditional parking systems often face challenges such as manual errors, time delays, and underutilization of parking spaces. An automated vehicle number plate detection system provides a robust solution by streamlining parking operations and minimizing human intervention.

1.2 Requirements

The development of an automated parking system involves addressing multiple conflicting requirements. These include achieving high accuracy in number plate detection while maintaining system affordability, ensuring compatibility between various hardware and software components, and managing real-time processing without compromising system performance.

1.3 Contribution

This project introduces a fully automated system that integrates EasyOCR and OpenCV for real-time number plate detection and slot allocation. Unlike conventional systems, this project emphasizes seamless hardware-software integration, efficient use of parking spaces, and user-friendly interaction through real-time updates displayed on an LCD.

Report Organization

This report is organized as follows:

- **Section 1: Introduction** – Discusses the project's motivation, scope, and structure.
- **Section 2: Objectives** – Outlines the primary goals of the system.
- **Section 3: Research Methodology and Implementation** – Details the system design, components, and working principles.
- **Section 4: Final Outcome Analysis** – Provides simulation results, performance evaluation, and challenges faced.
- **Section 5**: **Discussion**- Discusses challenges and Errors .
- **Section 6: Work Timeline** – Presents a Gantt chart and task schedule.

- **Section 7: Cost Analysis** – Includes a detailed breakdown of expenses.
- **Section 8: Impact on Society and Environment** – Evaluates the system's broader implications.
- **Section 9: Addressing Complex Engineering Problems and Activities** – Highlights engineering challenges and solutions.
- **Section 10: Conclusion** – Summarizes findings and suggests future improvements.
- **Section 11: References**

Section 2: Objectives

- To develop a fully automated system for real-time vehicle number plate detection and parking slot allocation.
- To achieve high accuracy in number plate recognition.
- To optimize the allocation and utilization of parking spaces.
- To provide real-time updates through an LCD interface.

Section 3:Research Methodology and Implementation

3.1 System Design Flowchart

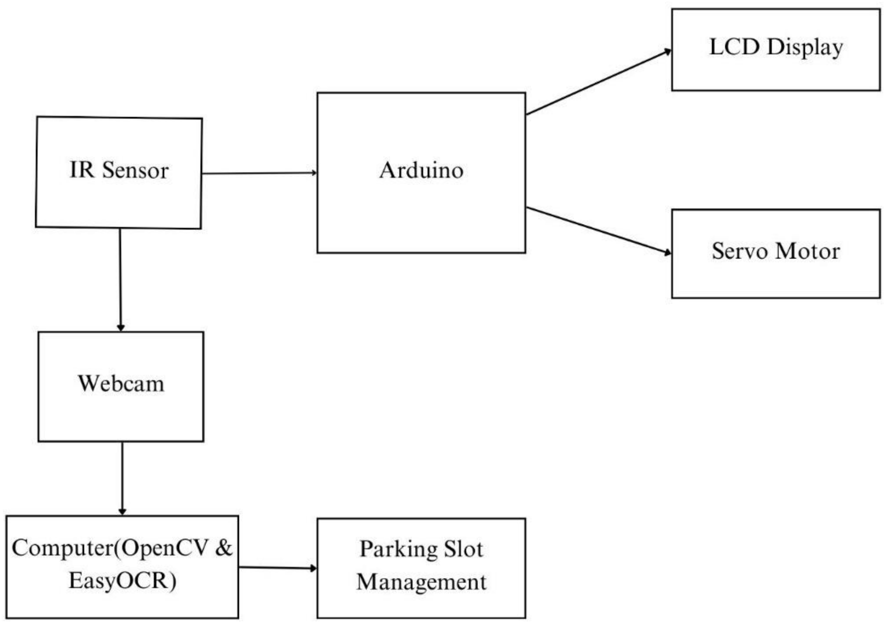

Fig-1: Block diagram of automated car parking system.

3.2 Circuit Diagram

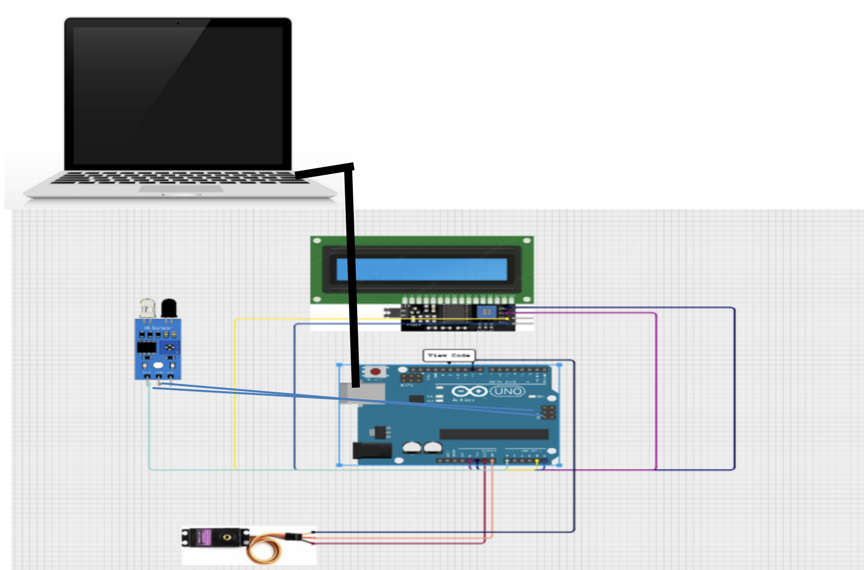

Fig 2: Circuit Diagram of Automated Car Parking System

Descriptions of all the components of Fig-1 is given below:

1. IR Sensor (Fig-3)

- Function: Detects the presence of a vehicle when it enters or exits the parking area.
- Output: Sends a signal to the Arduino Uno, indicating whether a vehicle has been detected.
- Purpose: Serves as the initial trigger for the entire system's workflow.

2. Arduino Uno (Fig-4)

- Function: Processes the signal received from the IR sensor and controls the LCD screen.
- Role: Acts as the intermediary between the sensor and the display system, ensuring that the appropriate information is sent to the LCD.
- Output: Communicates the sensor's status and slot updates to the LCD.

3. LCD Screen (Fig-5)

Function: Displays real-time parking slot statuses and provides recommendations to guide vehicles to available slots.

- Purpose: Provides clear visual feedback to users, making the parking process more efficient and user-friendly.

4. Webcam

- Function: Captures an image of the vehicle's number plate when triggered by the IR sensor.
- Output: Sends the captured image to the connected computer for processing.
- Purpose: Ensures accurate and efficient data collection for identifying vehicles.

5. Computer (OpenCV & EasyOCR)

- Function:

- Uses OpenCV to preprocess the image for better clarity and recognition.
- Utilizes EasyOCR to extract text (vehicle number plate) from the processed image.

- Output: Sends the recognized number plate information to the slot management system.
- Purpose: Serves as the brain of the system by analyzing and interpreting the captured data.

6. Parking Slot Management

- Function: Maintains and updates the status of parking slots.
    - Allocates an available slot to an incoming vehicle.
    - Frees up a slot when a vehicle exits.
- Output: Sends slot information back to the Arduino Uno for display on the LCD.
- Purpose: Optimizes parking space usage by efficiently tracking and managing available slots.

3.2 **Circuit Diagram**

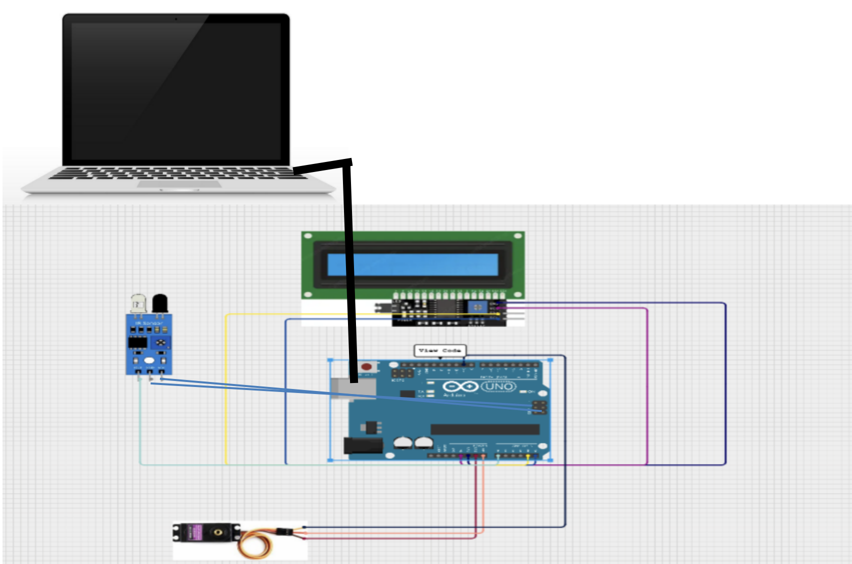

Fig 2: Circuit Diagram of Automated Car Parking System

3.3 Hardware Components

- **IR Sensor (Fig-3) :** An infrared sensor detects the presence of objects within its range by emitting infrared light and measuring the reflected light. It is widely used in applications such as proximity sensing and object detection. The sensor works effectively in various environmental conditions and offers high reliability and low power consumption.

Fig-3: IR Sensor

- **Arduino Uno (Fig-4) :** Arduino Uno is an open-source microcontroller board based on the ATmega328P chip. It offers an easy-to-use interface and supports a wide range of peripherals and modules. Arduino Uno is commonly used in embedded systems projects

  due to its simplicity, versatility, and extensive community support.

Fig-4: Arduino

- **LCD Screen (Fig-5) :** Liquid Crystal Displays (LCDs) are electronic display modules that use liquid crystal to produce visual output. They are energy-efficient, offer clear visibility, and come in various configurations such as 16x2 or 20x4, making them suitable for displaying short messages and system status updates in embedded projects.

Fig-5: LCD display

- **`Servo Motor (Fig-6)**: A servo motor is a rotary actuator designed for precise control of angular or linear position. It consists of a motor, a feedback sensor, and control circuitry. Servo motors are widely used in robotics and automation to achieve accurate movements. They receive control signals (typically PWM) to position the motor shaft at desired angles.

Fig-6: Servo Motor

3.4 Software Components

- **Python:** Python is a high-level programming language known for its simplicity and extensive libraries. It is widely used in fields like data analysis, machine learning, and computer vision. Python's versatility and integration capabilities make it an ideal choice for image processing and OCR applications.

- **OpenCV:** Open Source Computer Vision Library (OpenCV) is a robust framework for image processing and computer vision tasks. It supports various functions, including object detection, image enhancement, and feature extraction. OpenCV is widely adopted in both academic and industrial domains.

- **Arduino IDE:** The Arduino Integrated Development Environment (IDE) is used to program Arduino boards. It provides a user-friendly interface to write, compile, and upload code. The IDE supports various libraries and facilitates real-time debugging, making it essential for prototyping and deploying embedded applications.

Section 4 :Result Analysis

1. Figure 7 illustrates the LCD output displaying the slot allocation for the second car. Similarly, all subsequent cars can view the available parking slots on the LCD display upon their arrival.

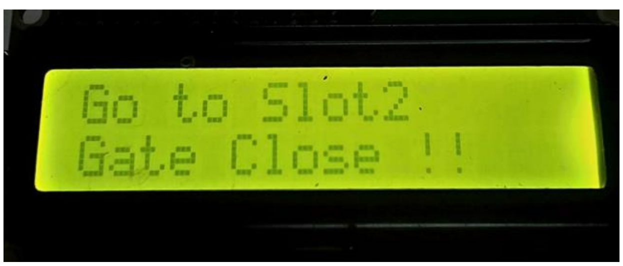

Fig 7: LCD Output.

2. In Figure 8, the user interface output shows the status of the parking lot when all slots are occupied. At this point, the system indicates that no slots are currently available.

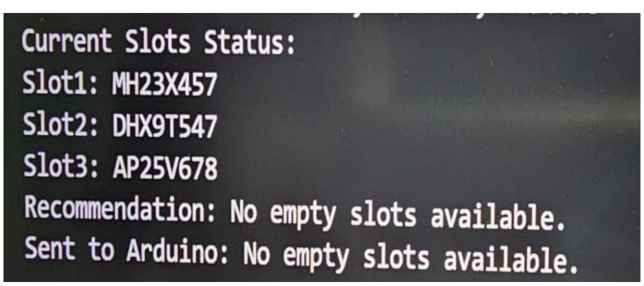

Fig 8: User Interface when no slots are empty.

3. In Figure 9, we observe the updated status after the second car exits the parking slot. The display shows that Slot 2 is now empty. Consequently, when a new car arrives at the parking lot, it will be directed to occupy Slot 2.

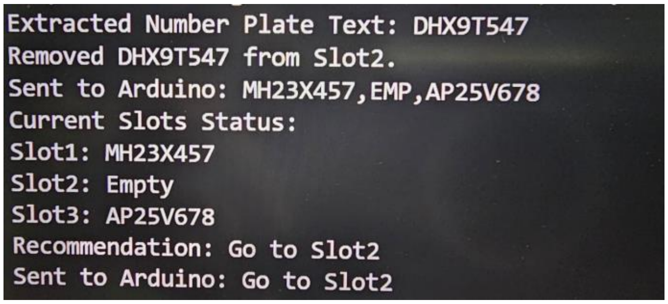

Fig 9: User interface when a car of slot 2 leaves.

Section 5: Discussion

During the development of the automated car parking system, several challenges were encountered that required careful troubleshooting and refinement. One major issue was the difficulty in recognising number plates for camera focusing, which affected the accuracy of the OCR system. To address this, the image preprocessing steps were optimized, but certain cases still presented limitations. Another challenge was synchronization between hardware components, particularly during initial testing, where delays and mismatches in sensor triggering and data processing were observed. This was mitigated by fine-tuning the timing and ensuring stable connections. Additionally, noise in serial communication caused interruptions in data transfer between the Arduino and the software system. Implementing error-checking mechanisms and optimizing the communication protocol helped to minimize these issues, ensuring smooth and reliable operation.

Section 6: Work Timeline

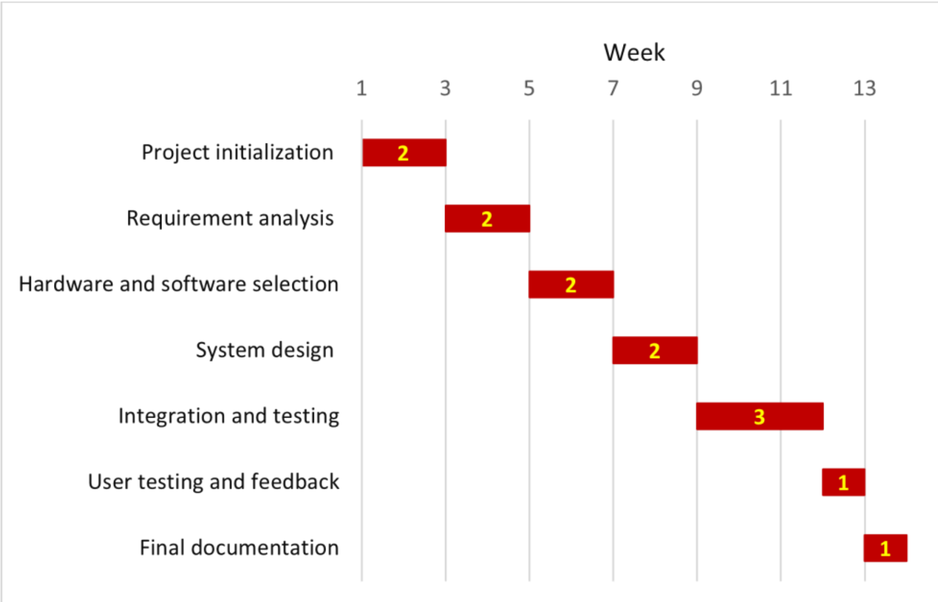

Fig-10: Work timeline (Gantt chart) of this project.

Section 7: Cost Analysis

Table-1: Necessary hardware components needed for the project with cost analysis:

Hardware	**Cost**
IR Sensor	70/-
Arduino	515/-
LCD Sensor	240/-
Miscellaneous(Connectors)	30/-
Servo Motor	135/-
Webcam	1000/-
Total Cost	**1990/-**

Section 8: Impact on Society and Environment

8.1 Impact on Society:

- Improves parking efficiency and reduces congestion in urban areas.
- Enhances user convenience and security through automatic number plate detection.

8.2 Impact on Environment:

- Reduces carbon emissions by minimizing idle vehicle time during parking.
- Promotes eco-friendly urban development by optimizing land use and reducing infrastructure expansion.

Section 9: Addressing Complex Engineering Problems and Activities

9.1 Addressing Complex Engineering Problems

Serial	Attributes	Addressisng complex engineering problems in the project
P1	Depth of knowledge required	• Requires understanding of microcontroller programming, serial communication, computer vision (OpenCV), optical character recognition (OCR), and hardware integration (IR sensors and LCDs). • Knowledge of Python libraries like EasyOCR for text recognition and concepts in embedded systems is essential.
P2	Range of Conflicting Requirements	• Balancing the need for accurate number plate detection with system speed and resource limitations. • Managing synchronization between hardware (IR sensor and Arduino) and software processing (OCR and slot management).
P3	Interdependence	• The project integrates several subsystems: IR sensor for car detection, webcam for OCR, and Arduino for hardware output.

9.2 Addressing Complex Engineering Activities

Serial	Attributes	Addressisng complex engineering problems in the project
A1	Range of Resources	Involves using a webcam, IR sensors, Arduino, Python-based OCR tools (EasyOCR), and software libraries like OpenCV for image processing.
A2	Level of Interactions	Requires integration of software (Python programs) with hardware (IR sensor, Arduino) to ensure seamless communication and real-time decision-making.
A3	Familiarity	Gained expertise in Python programming, hardware-software integration, image preprocessing, and system optimization techniques.

Section 10: Conclusion

The automated vehicle number plate detection system demonstrates significant potential for modern parking management. This system not only improves user convenience but also reduces congestion and environmental impact, making it a practical contribution to smart city initiatives. Future enhancements could include integrating cloud-based data management, extending the number of slots, and improving recognition accuracy under challenging conditions, further expanding the system's scalability and reliability.

Section 11: References

1.https://www.ijraset.com/research-paper/smart-parking-system-using-python-and-opencv